

V. HARSHA

B. Tech Artificial Intelligence and Machine Learning
Computer Vision | Generative AI | Autonomous Systems
Chennai, India
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PROFESSIONAL SUMMARY

- B.Tech AI-ML graduate specializing in **Computer Vision, Generative AI, and Intelligent Automation Systems**.
- Hands-on experience building real-time AI systems, including **ADAS, multi-agent LLM frameworks, and object detection pipelines**.
- Strong technical background in **Python, TensorFlow, OpenCV, and LangChain**, with research experience in **deep learning and AI-driven systems**.
- Interested in **Research & Development** in Artificial Intelligence, Computer Vision, Autonomous Systems, and Intelligent Automation.

EDUCATION

- **Bachelor of Artificial Intelligence and Machine Learning** (2022-2026) from Sri Ramachandra faculty of engineering and technology, Chennai, India - **CGPA: 8.46/10**
- **High School** - Sri Chaitanya Techno School, Chennai, India (10th and 12th Grade)

HONOURS

Second Prize (State Level) – English Elocution Competition, World Space Week 2020, conducted by **Satish Dhawan Space Centre (SDSC SHAR), ISRO**.

TECHNICAL SKILLS

- **Programming:** Python, Java, C, SQL
- **AI/ML & Deep Learning:** Machine Learning, Deep Learning, CNN, TensorFlow, PyTorch, Scikit-learn
- **Computer Vision:** OpenCV, YOLO, Object Detection, Image Processing, Facial Expression Recognition
- **Generative AI & NLP:** LLMs, LangChain, RAG, Multi-Agent AI Systems, Prompt Engineering
- **Development Tools:** GitHub, VS Code, Google Colab, Jupyter Notebook, n8n
- **Data & Analytics:** Data Analysis, Data Visualization, Data Preprocessing
- **APIs & Automation:** OpenAI API, Groq API, Telegram API, Voice & Speech-Based Systems

INTERNSHIPS & EXPERIENCE

Hexaware Technologies Limited
Gen AI Intern

Feb 2026 – Jun 2026
Chennai, India

- Developed a Retrieval-Augmented Generation (RAG) based documentation assistant using Large Language Models (LLMs), vector embeddings, and retrieval pipelines.
- Applied prompt engineering and LLM integration techniques for intelligent information retrieval and response generation.
- Explored Reinforcement Learning concepts for optimizing AI model behavior and decision-making processes.
- Designed and implemented a multi-Agent AI system for automated decision-making, enabling collaborative task execution through orchestrated LLM workflows.

Eubix Technologies Pvt Ltd, IITM Research Park
ML Intern

Aug 2024 - Oct 2024
Chennai, India

ADVANCED DRIVER ASSISTANCE SYSTEM (ADAS) USING COMPUTER VISION

- Developed a real-time Advanced Driver Assistance System (ADAS) for driver monitoring, vehicle safety, and road hazard detection.
- Implemented drowsiness and distraction detection using facial landmark analysis, Eye Aspect Ratio (EAR), and Mouth Aspect Ratio (MAR).
- Developed lane detection and lane departure warning modules using image processing and computer vision techniques.
- Built safety monitoring features including seatbelt detection, mobile phone usage detection, and smoke detection.
- Integrated road hazard detection for identifying potholes and unsafe road conditions in real time.

NEXT 24TECH
AI-ML Intern

Feb 2024 - Apr 2024
Maharashtra, India

- Developed an image classification model using Convolutional Neural Networks (CNNs) on the CIFAR-10 dataset.
- Performed data preprocessing, augmentation, training, and evaluation to improve model accuracy and generalization.
- Utilized TensorFlow and Keras for deep learning model development and optimization.
- Gained hands-on experience in deep learning workflows, including model training, validation, and performance tuning.

Internal Internship (Sri Ramachandra Faculty of Engineering and Technology)
Web Development Intern

Apr 2023 – Jun 2023
Chennai, India

- Developed a full-stack web application for an online food delivery system.
- Designed responsive user interfaces with menu browsing, cuisine selection, and cart management features.
- Implemented basic backend functionality and database integration for user and order management.
- Contributed to application design, testing, and deployment activities.

PUBLICATIONS & RESEARCH

[Advanced Driver Assistance System \(ADAS\) – SSRN \(2025\)](#)

- Developed a real-time ADAS integrating computer vision and machine learning for driver monitoring, lane detection, object detection, and road hazard identification. Implemented safety features including drowsiness detection, seatbelt detection, phone usage monitoring, and real-time alert generation.

Deep Multi-Task Learning for Simultaneous Emotion Recognition and Activity Monitoring using Wearable Sensors – ZENODO (2025)

- Proposed a CNN-based multi-task learning framework for simultaneous emotion recognition and physical activity monitoring using wearable sensor data. Evaluated the model on benchmark datasets to improve accuracy and computational efficiency.

Small Language Models (SLMs) in Autonomous AI Agents for Healthcare – Manuscript (In Progress)

Research focused on agentic AI systems, reinforcement learning, and healthcare decision-support automation using SLMs.

PROJECTS

- **Campus Security System**

Built a real-time surveillance system using OpenCV and YOLO for detecting suspicious activities in monitored environments. The system performs automated video analysis, object detection, and real-time alert generation to enhance security monitoring and threat detection capabilities.

- **Automation Telegram Bot (n8n + AI)**

Developed an intelligent automation bot using n8n workflow automation and Telegram API to execute tasks through natural language commands. Integrated AI capabilities to interpret user inputs and convert them into automated workflow executions, improving task automation efficiency and usability.

- **AI-Powered Multimodal Desktop Automation System (Ongoing)**

Currently developing an advanced multimodal AI desktop assistant that combines voice control, vision-based automation, and intelligent task execution. The system integrates AI models for natural language understanding, computer vision-based screen interaction, and automation control to perform end-to-end desktop operations. The goal is to create a unified AI assistant capable of acting as a personal productivity agent with multimodal input handling (text, voice, and visual cues).

CERTIFICATIONS

- **IBM - Generative AI Essentials Using LLMs**
- **AWS Academy – Introduction to Cloud Computing**
- **Accenture (Forage) – Data Analysis and Visualization, Frontend & Backend Development (Job Simulation)**
- **Kaggle – Python Programming Certification**
- **HackerRank – Java Programming, Problem Solving, Software Engineering Fundamentals**
- **Edureka – Java Programming Certification**
- **Tableau – Data Analysis & Visualization Certification**

LANGUAGES

- **English** – Professional Working Proficiency
- **Hindi** – Professional working Proficiency
- **Japanese** – NAT 5Q Certified | JLPT N3 Aspirant
- **Tamil** – Native Proficiency